

Younsoo Park

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Education

Pennsylvania State University, University Park, PA
B.S. in Computer Science (Minor in Mathematics)

Aug 2021 – Present
Expected Graduation: May 2027

Experience

Levit Inc. (Shoppoort AI) - Associate Problem Solver

June 2026 – Aug 2026

- Worked directly with the **CEO** on Shoppoort, Levit's live iOS/Android AI shopping app, across routing, evaluation, and personalization systems.
- Evolved routing from LLM classification to an embedding-based semantic router (**$\sim 8\times$ faster**) and a hybrid classifier, reaching **86%** V3 accuracy and rescuing **93%** of eligible errors.
- Designed and shipped Wowness, a Mem0-backed cross-category recommender, as sole engineer, and built the production evaluation & backfill platform end-to-end.
- Diagnosed system-boundary production failures and corrected analytics false positives; contributed **730 commits** and **143 merged PRs** (+114K LOC).

SIGNUM (Nittany AI Alliance – PIT-UN Funded)

Aug 2024 – Nov 2025

Co-founded an AI-powered caregiver support platform that received \$3,750 in PIT-UN funding.

- Built a HIPAA-aware RAG assistant using AWS Bedrock, FAISS, and spaCy.
- Integrated CMS, NPPES, and Google Places APIs, cutting caregiver research time from 50+ hrs/week to near-instant.

Penn State Advanced Vehicle Team (EcoCAR) - Project Manager

Aug 2026 – Present

- Manage planning and deliverables for Penn State's EcoCAR team (Argonne National Laboratory, GM, MathWorks).
- Building Year 1 structure and processes for future teams while keeping vehicle work on schedule.

Atom Tech Solutions LTD, Internship — Berkeley, CA

May 2022 – Aug 2022

- Developed secure login system with SQL and JWT based authentication.
- Implemented encrypted data storage and credential generation features.

Projects

Undergraduate Research Assistant & Team Lead

Oct 2025 – Present

- Researched federated and adversarially robust TinyML for IoT security, improving attack recall from 46.7% to 93.9% and F1 score from 84.1% to 89.3%.
- Compressed deployment models by 12.3 $\times$ and reduced ESP32 inference latency by 74.5% across 48 evaluated configurations using pruning, knowledge distillation, and INT8 quantization.

AI Agent Memory Systems

Jul 2026 – Present

Researched 5 agent memory frameworks (Mem0, A-MEM, LangMem, Zep, OpenAI Memory) to identify the SOTA approach, under Prof. Kiwan Maeng.

XIIO – LinkedIn for Filmmakers

Apr 2026 – Present

Building a LinkedIn-style platform for filmmakers with creator portfolios, video streaming, and collaboration using Next.js, Firebase, and Cloudflare Stream.

ASME @ Penn State Website

Aug 2024 – Present

Built the ASME Penn State website with React, TypeScript, and Firebase, with real-time Firestore listeners and role-based approval workflows.

Skills

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL

Frameworks: React, Next.js, Node.js, NestJS, Express, Flask, Firebase

AI/ML: PyTorch, TensorFlow, Flower, scikit-learn, spaCy, FAISS, Mem0, AWS Bedrock, OpenAI API

Data & DevOps: PostgreSQL, MongoDB, Redis, Docker, Kubernetes, Helm, GitHub Actions, Datadog

Awards & Honors

- 1st Place, ThinkNeuro LLC Project Competition Apr 2026
- 1st Place, IEEE Student Chapter Battle Bots Competition Apr 2026
- PIT-UN Funding Award (\$3,750) – Nittany AI Challenge Apr 2025
- 3rd Place, ACM Pathfinder Robotics Challenge Apr 2025
- Dean's List, Penn State August 2021 - Present

Activities

Penn State THON™, Technology Captain

Apr 2026 – Present

Republic of Korea Air Force, Squad Leader, led 20+ airmen (Commendation Award)

Sept 2022 – June 2024

ACM, Web & Design Director

Apr 2026 – Present

Lead Teaching Assistant, Math 230 (Multivariable Calculus)

Aug 2025 – Present